Lec-1

UNIT-1

* Overview of Soil Mechanics *

* Soil Mechanics :-

- It is a branch of science that deals with the study of physical property of soil masses subjected to various types of loads.

- Karl Terzaghi is known as the father of Soil mechanics.

- According to Terzaghi :- Soil mechanics is the application of civil engineering involving the study of soil, its behaviour and application as the engineering material.

* Soil :-

→ It has different meaning according to different conditions.

→ Common man's definition :- It is the upper surface of the earth which contains loose materials such as clay, Silt, sand and gravel etc.

→ Agriculturists definition :- It is the part of earth surface which supports plant life.

→ Engineer's definition :- It is the mixture of minerals and rocks, obtained from chemical and mechanical weathering of rock.

→ As per IS (2809-1972) :- It is the sediment or other consolidated accumulation of solid particles produced by physical and chemical disintegration of rock.

* Importance of Soil in Civil Engineering :-

use of soil

→ Soil is used as either construction material or as a foundation support (bed) for structures.

A. As construction material

→ Due to its usefulness, easy availability and low cost it is a widely used material.

→ It is used for housing, road and railway embankment etc.

→ It is used for brick manufacturing & these bricks are used for building construction.

→ Pervious and impervious soil can be used in earthen dams.

→ For plinth filling, soil can be used as construction material.

→ As a Sub grade material for road construction.

→ In retaining, soil is used as filling material.

* Origin of Soil :-

→ Soil is formed by disintegration of rock either by physical or chemical weathering.

Stages in geological cycle of soil formation :-

⇒ Weathering (I)

⇒ Transportation (II)

⇒ Deposition (III) → Layer-by-Layer

⇒ Upheaval (IV)

- Soil formation is a cycle called 'Geological Cycle'.
- Process of soil formation is called 'Pedogenesis'.

Classification of Soil :-

I) Residual Soil
II) Transported Soil

I) <u>Residual Soil</u> :- Soil that remain at the location of its formation are called residual soil.

II) <u>Transported Soil</u> :- Soil that are transported from its place of origin by wind, water or glacier are called transported soil.

<u>Note</u> :- Residual soil have better engineering properties as compared to the transported soil.

According to transporting agencies, soil are classified as follows :-

⇒ Alluvial Soil :- Deposite from suspension in running water/river.

⇒ Lacustrine Soil :- Deposite by still water or lakes.

⇒ Marine Soil :- Deposite by sea water.

⇒ Aeolian Soil :- Deposite by wind. E.g. - loess

⇒ Glacial Soil :- Transported by ice.

⇒ Colluvial Soil :- Transported by gravity.

Some Special Soils :-

⇒ Bentonite :- obtained from the decomposition of volcanic ash & is highly plastic in nature.

⇒ Black cotton soil :- It is a residual soil formed from the basalt or trap rocks. It has. It has low bearing capacity, and high swelling and shrinkage characteristic.

⇒ Loam :- It is mixture of sand, silt and clay.

⇒ Talus :- Soil transported by gravity.

⇒ Marl :- A very fine-grained calcium carbonate soil of marine origin.

⇒ Organic Soil :-

- It is formed by the growth and subsequent decomposition of plants.
- Highly compressiable and are not suitable for engineering purpose.
- Eg. - muck, peat, humus.

⇒ Loam soil is also called 'Garden Soils'.

⇒ Organic soil is also called 'Cumulous Soils'.

* Field application of Geotechnical engineering :-

Some important applications of Geo-technical engineering :-

⇒ Foundation Design

⇒ Pavement Design

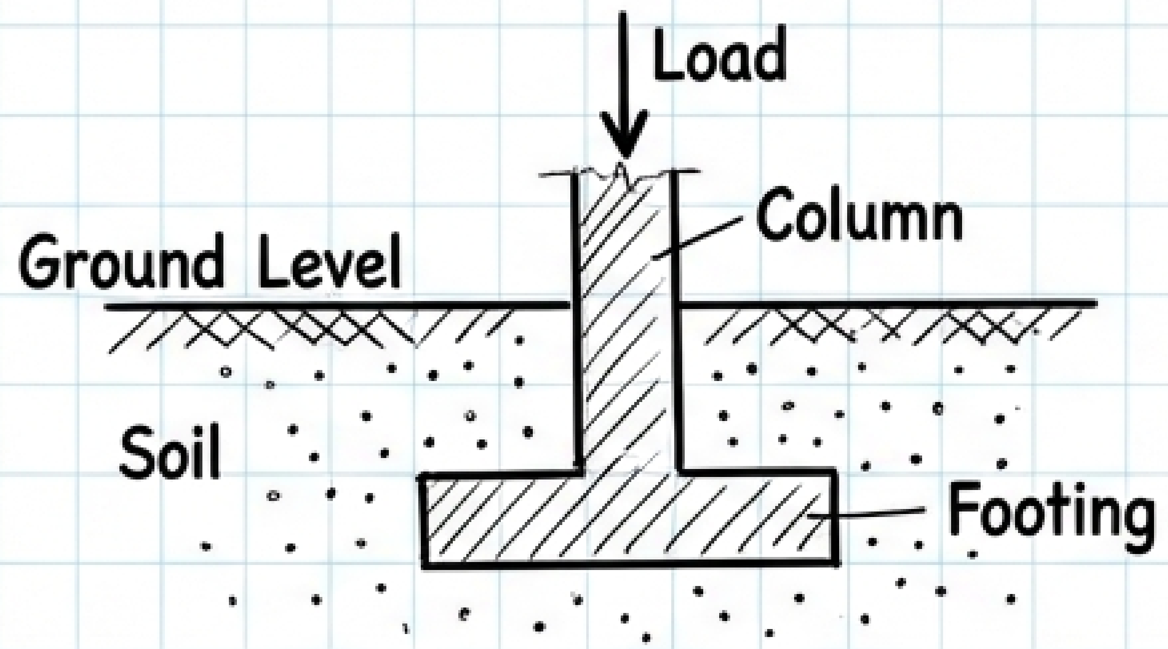
⇒ Design of earth retaining structures

⇒ Design of embankment

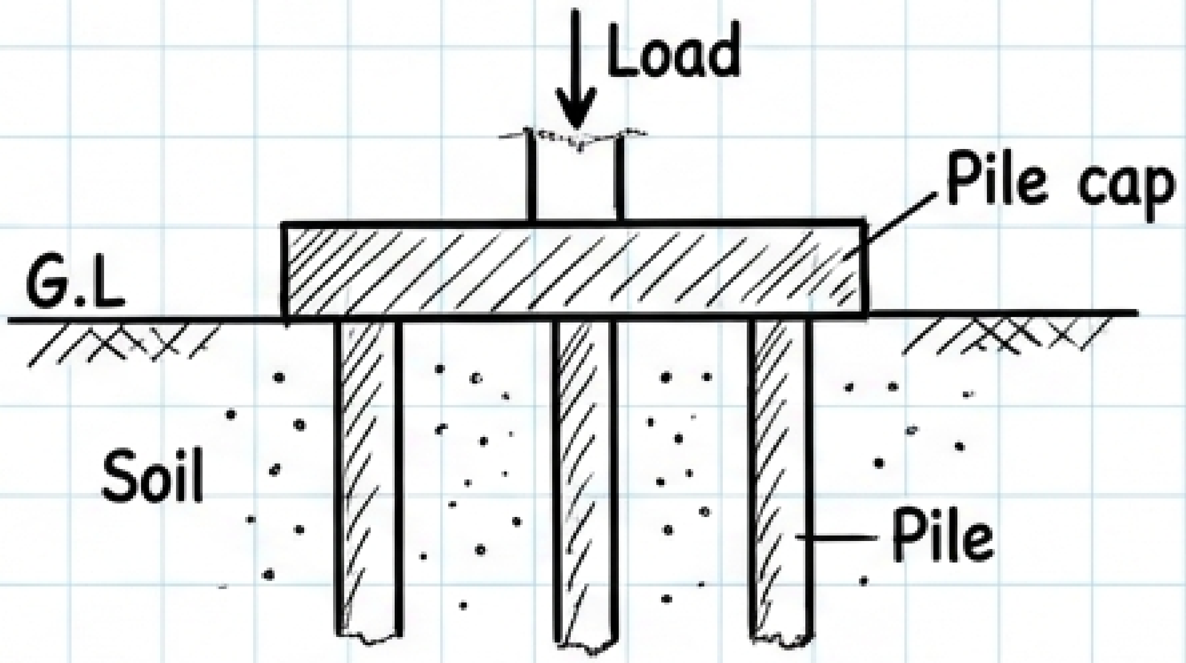
⇒ Underground structures

* Foundation Design :-

- Every structures (such as building, bridge, highway or dam etc.) lies in or on the earth surface.
- Foundation is required to transfer the load of structure to the soil safely and efficiently.
- Thus, it is important to know the bearing capacity of soil, knowledge of stress distribution below the loaded area, settlement of foundation, effect of vibration, effect of ground water etc.



(a) Shallow Foundation



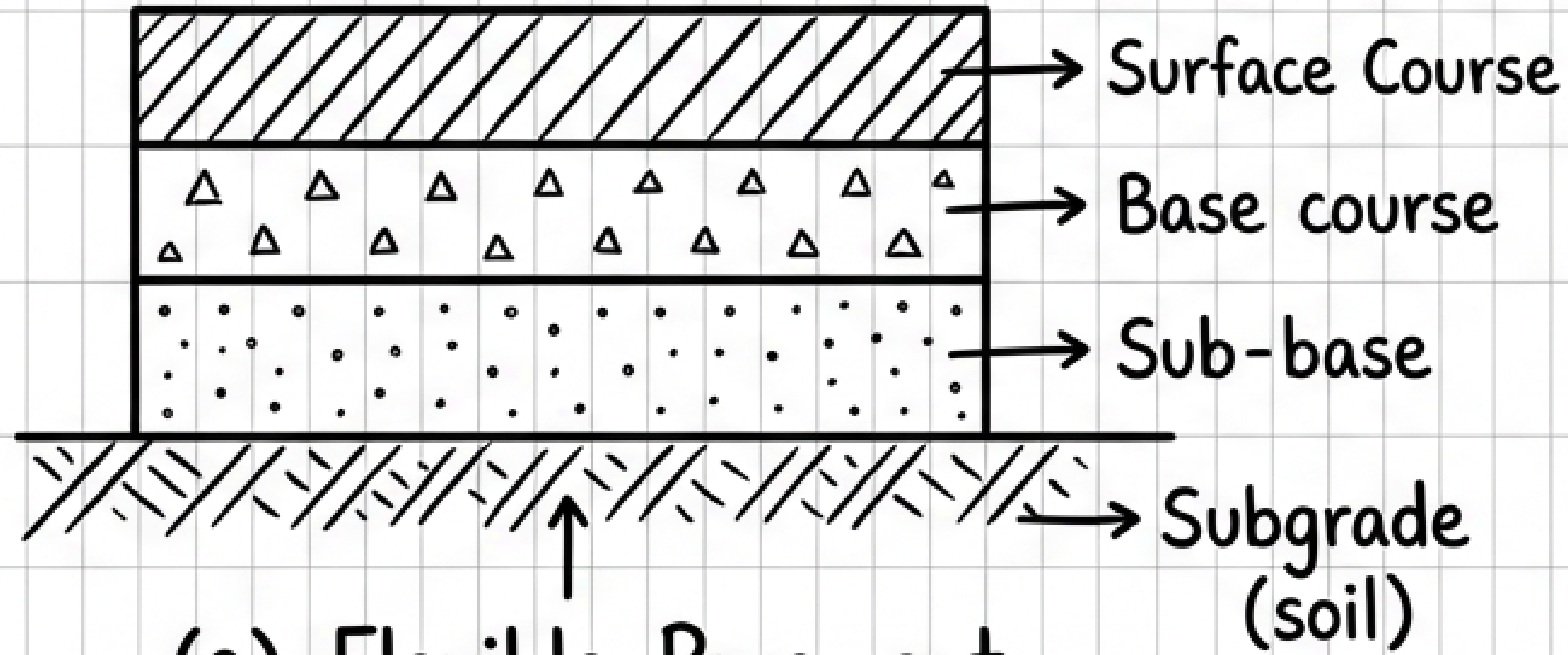
(b) Deep Foundation (Pile foundation)

Pavement Design :-

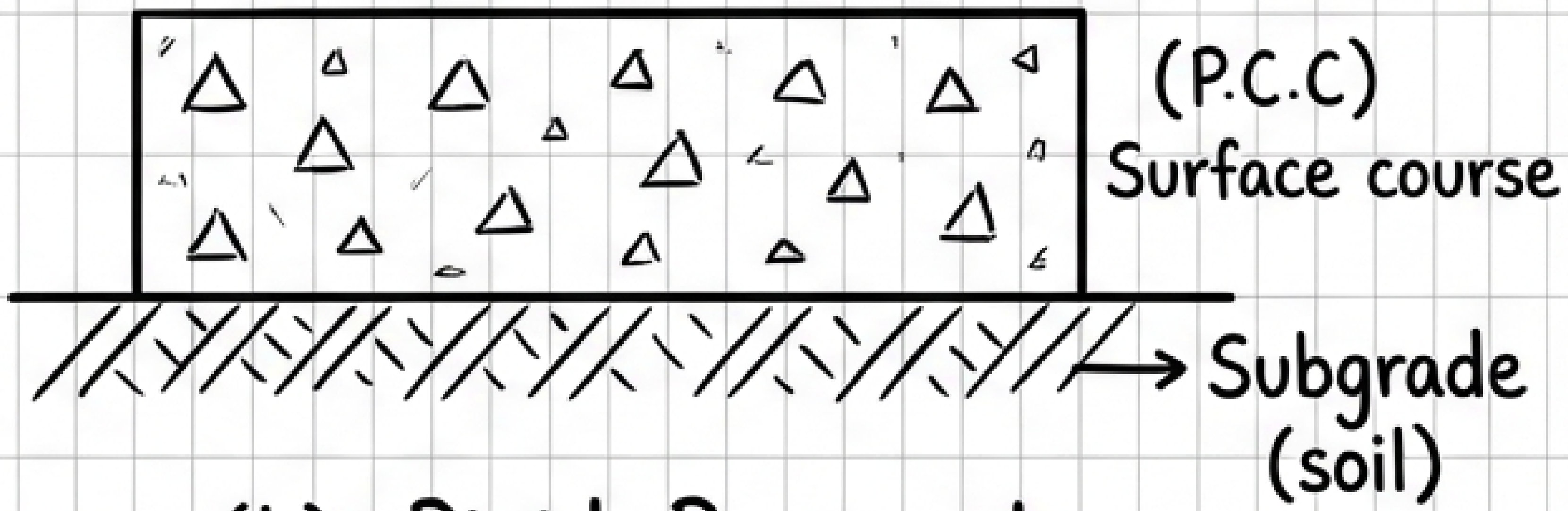
→ Pavement are provided on soil (Sub-grade) for the purpose of providing a smooth and strong surface for vehicle movement.

→ Pavement are of two types :- Flexible & Rigid.

→ Pavement thickness depends on subsoil, repetition of loading, intensity of traffic, construction materials, earth fills etc.



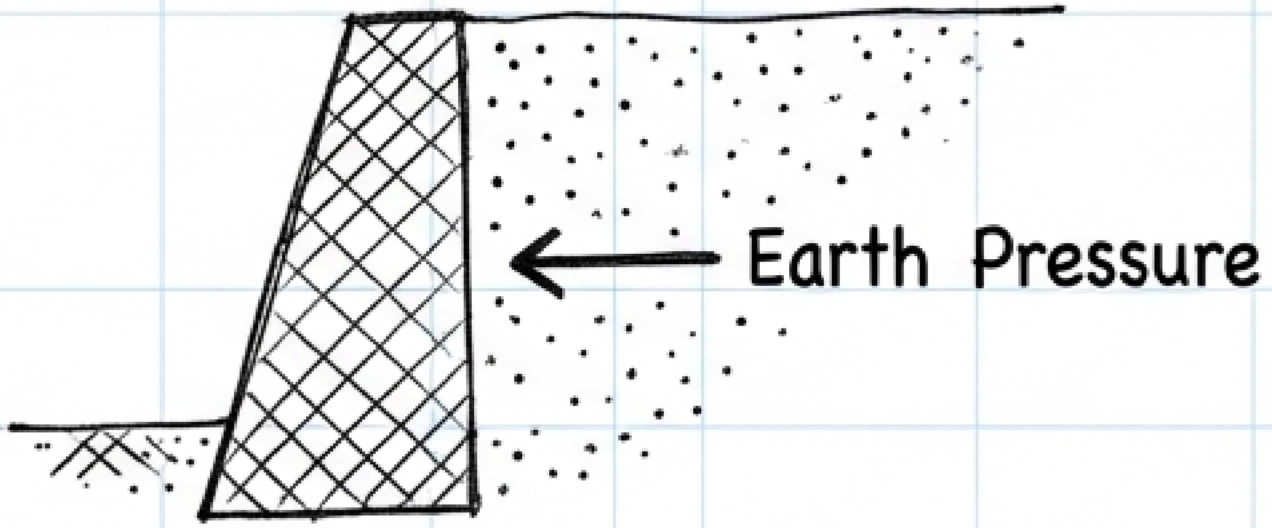
(a) Flexible Pavement



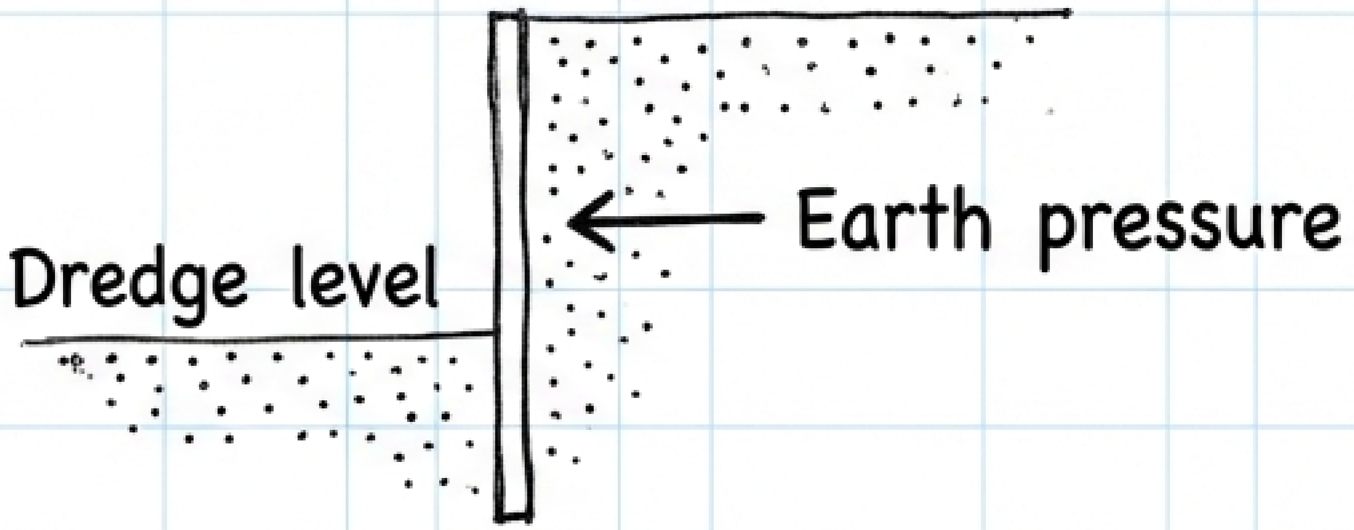
(b) Rigid Pavement

* Design of earth retaining Structure * :-

- When sufficient space is not available for soil mass to spread & form a slope, a structure is required to retain the soil.
- Earth retaining structure is also required to keep the soil at different levels on its either side.
- The retaining structure may be a rigid retaining wall or sheet pile (relatively flexible).
- The knowledge of the active earth pressure, passive earth pressure, density & moisture content is essential for design of earth retaining structures.



(a) Retaining wall

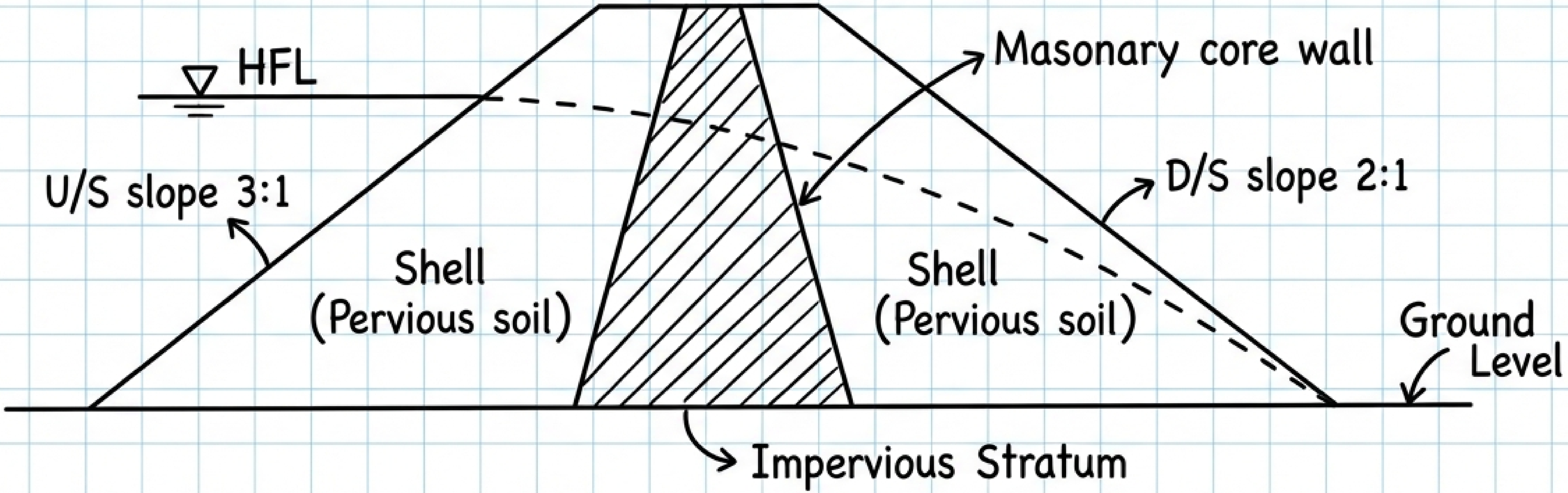


(b) Sheet Pile

* Design of Earthen dam *

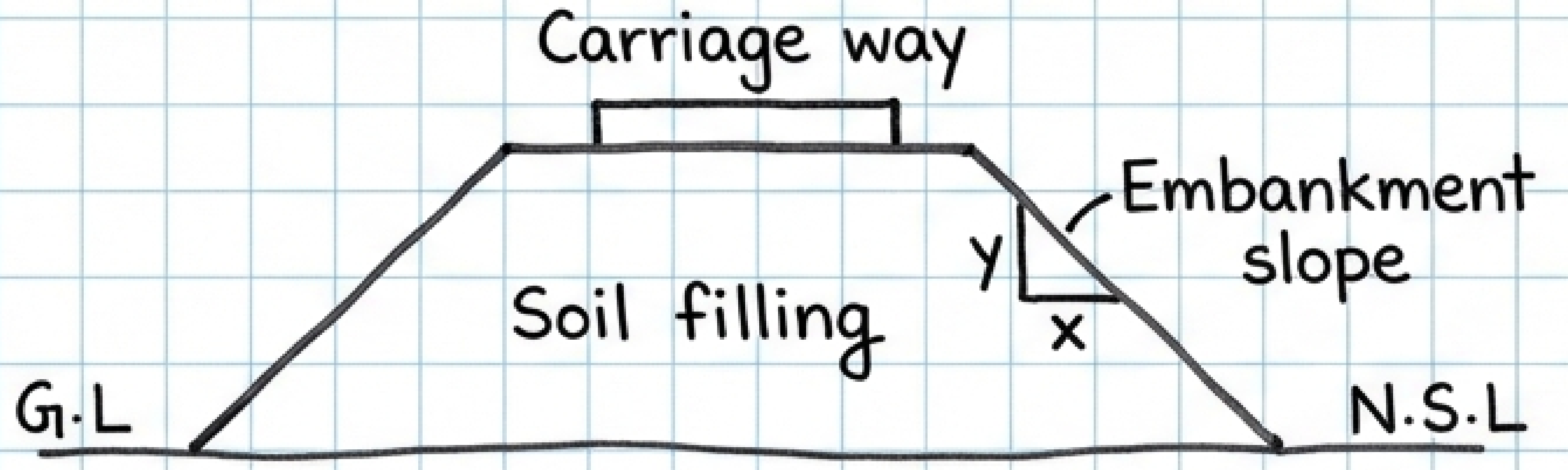
- Large amount of pervious and impervious soil is required for earthen dam construction.
- Its design requires knowledge of index properties, Plasticity characteristics, particle size distribution specific gravity, permeability, compaction, shear strength & bearing capacity etc.

* Earthen Dam *

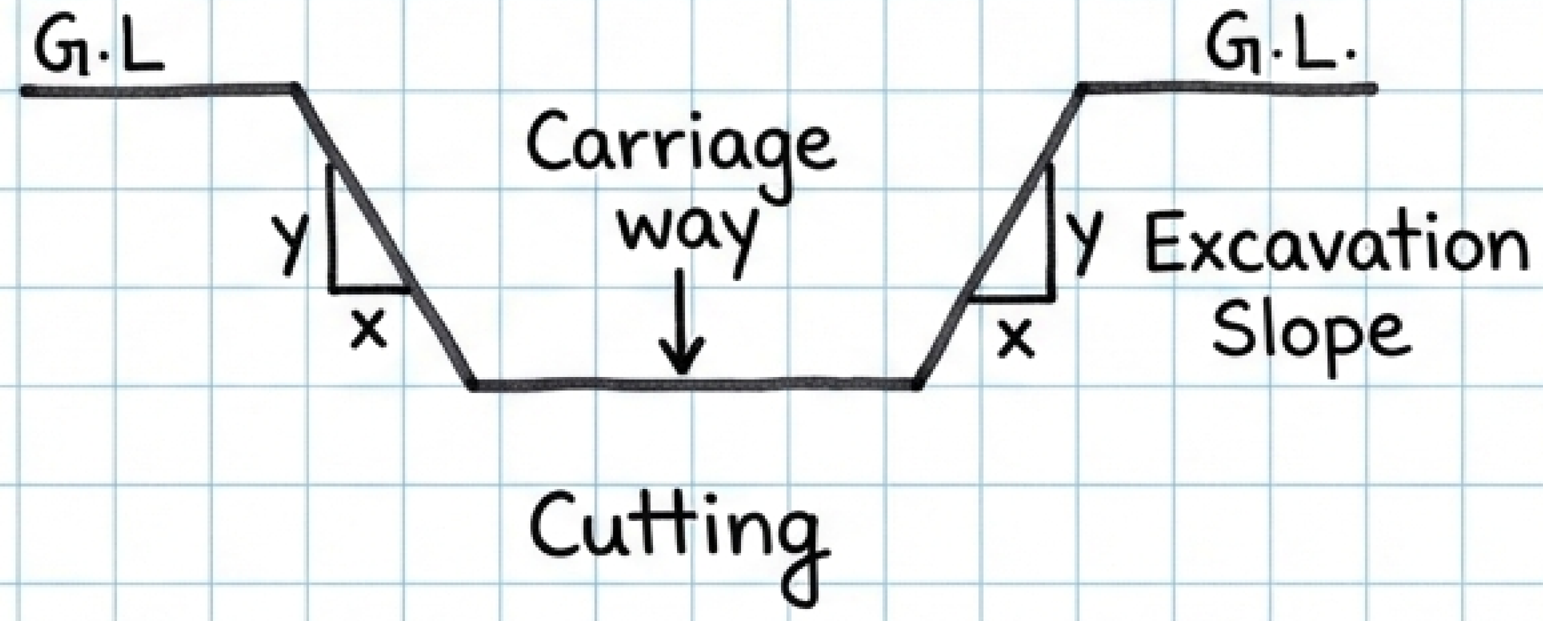


* Design of embankment *

- For stable design of slopes of filling and cutting, the knowledge of shear strength, angle of repose and frictional coefficient is necessary.



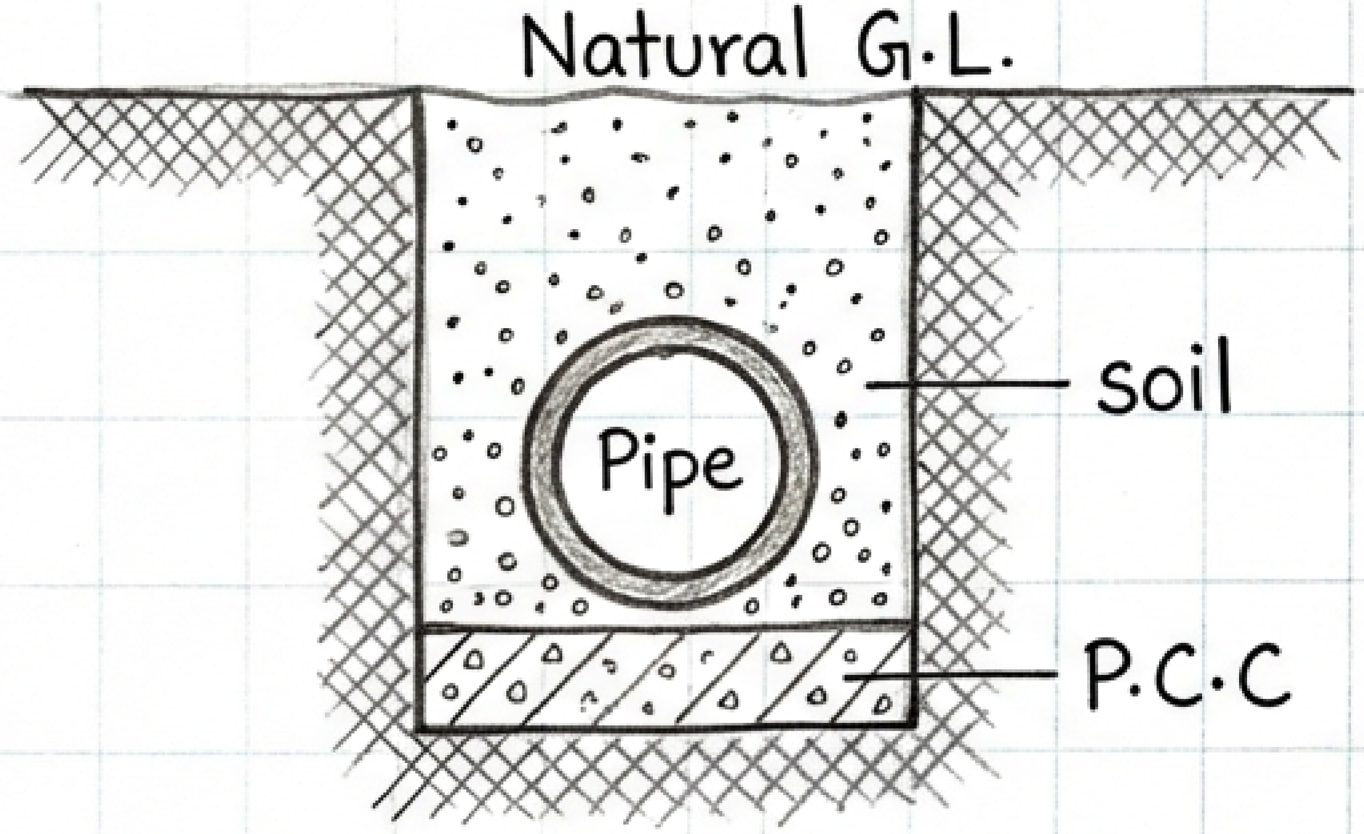
(a) Slope in filling for Road



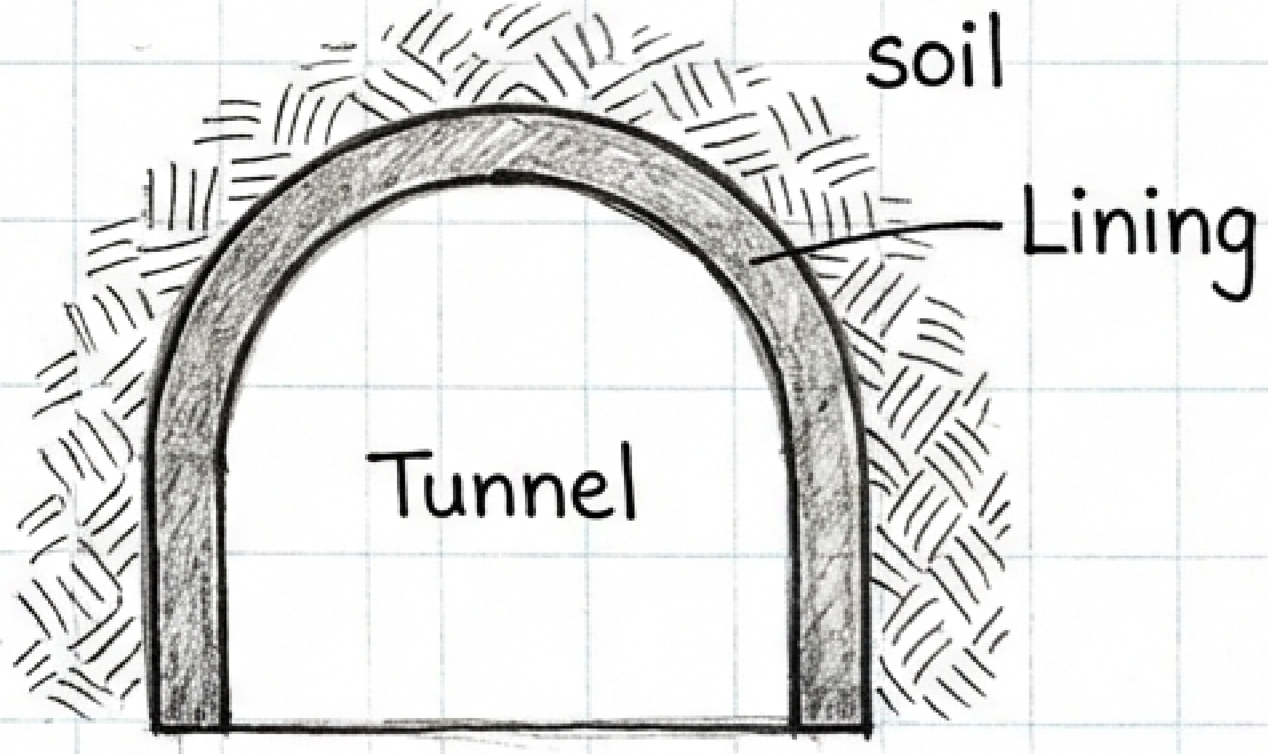
(b) Slope in cutting for Road

* under-ground structures *

Design & construction of underground structures (such as pipe line, tunnels, underground buildings etc.) requires the knowledge of density of soil, consolidation of soil, bearing capacity of soil & water condition of sub-soil layers in necessary.



(a) Pipe Line
(underground)



(b) Tunnel

* Objectives of Geotechnical engineering * :-

Following are the objectives of Geotechnical engineering :-

⇒ To perform Soil investigation

⇒ To develope method for Soil Sampling

⇒ To classify soil properties regarding engineering properties of Soil.

⇒ To apply the results of soil investigation & sampling, to use the soil as construction material.